

FEATURE ENGINEERING DOCUMENT (FED) v2.0

AI vs Human Content Detection System

Project Name: AI vs Human Content Detection System

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Status: Approved for Client Handover

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DOCUMENT CONTROL

Version	Date	Author	Description
1.0		Project Initiation Data Science Team	Initial Feature Engineering Plan
2.0	June 2026	M. Sagar Reddy	Final Implemented Feature Engineering Framework

APPROVAL MATRIX

Role	Responsibility	Status
Client Stakeholder	Review	Pending
Data Science Lead	Technical Approval	Completed
Deployment Team	Awareness	Completed

1. PURPOSE

The purpose of this document is to describe the complete feature engineering lifecycle followed during the development of the AI vs Human Content Detection System.

This document details:

- Raw features received from the client
- Engineered features created during development
- Feature selection decisions
- Feature exclusion decisions
- Leakage investigation findings
- Final production feature architecture

The document serves as the official reference for all feature-related decisions implemented in the final machine learning solution.

2. FEATURE ENGINEERING OBJECTIVES

The feature engineering process was designed to achieve the following objectives:

Business Objectives

- Improve AI content detection accuracy
 - Preserve human-authored content
 - Support cross-domain classification
 - Improve generalization capability
-

Technical Objectives

- Reduce feature leakage
 - Improve model stability
 - Improve external validation performance
 - Enable production deployment
 - Minimize inference complexity
-

3. ORIGINAL DATASET FEATURES

The client dataset contained multiple attributes describing content characteristics.

Raw Dataset Features

Feature	Description
text_content	Original textual content
word_count	Number of words
avg_sentence_length	Average sentence length
domain	Academic, News, Social
topic_hint	Content topic
source_model	AI model source
generation_method	Generation metadata
label	Target variable

4. INITIAL FEATURE ENGINEERING STRATEGY

The original feature engineering plan included both:

Structured Features

- Word Count
- Average Sentence Length
- Domain
- Topic Information

NLP Features

- TF-IDF Features
- N-Grams
- Text Pattern Features

The goal was to evaluate whether combining structured and textual features would improve performance.

5. EXPLORATORY FEATURE ANALYSIS

During exploratory data analysis, several observations were identified.

Observation 1

AI-generated content generally exhibited:

- More formal structure
 - Consistent sentence construction
 - Repetitive phrasing patterns
-

Observation 2

Human-written content frequently exhibited:

- Informal language
 - Personal experiences
 - Variable sentence structures
 - Inconsistent punctuation
-

Observation 3

Certain metadata fields appeared highly predictive and required further investigation.

6. LEAKAGE-PRONE FEATURES IDENTIFIED

A dedicated feature review identified several features that posed deployment or leakage risks.

source_model

Description

Identifies the AI model used to generate content.

Example

GPT

Gemini

Claude

Risk

Directly exposes content origin.

Decision

Removed from production model.

generation_method

Description

Contains generation metadata.

Risk

Unavailable during real-world inference.

Decision

Removed.

topic_hint

Description

Content category metadata.

Risk

Not guaranteed during production prediction.

Decision

Removed.

domain

Description

Academic, News, Social.

Risk

May not always be supplied during inference.

Decision

Excluded from final model.

7. ENGINEERED STRUCTURED FEATURES

Several handcrafted features were engineered during experimentation.

Word Count

Formula

```
len(text.split())
```

Purpose:

Measure content length.

Average Sentence Length

Formula

```
sum(words_per_sentence) / number_of_sentences
```

Purpose:

Measure writing complexity.

Observation

The feature provided useful exploratory insights but contributed limited production value compared to NLP features.

8. TEXT PREPROCESSING FEATURES

Before feature extraction, all text passed through a cleaning pipeline.

Lowercase Conversion

Example:

THIS STUDY INVESTIGATES AI

↓

this study investigates ai

Special Character Removal

Example:

AI!!! @@@ Study???

↓

ai study

Whitespace Normalization

Example:

ai study

↓

ai study

9. WORD-LEVEL TF-IDF FEATURES

The final solution incorporates Word-Level TF-IDF representations.

Objective

Capture:

- Vocabulary usage
 - Common phrases
 - Domain-specific terminology
-

Configuration

```
TfidfVectorizer(  
    stop_words='english',  
    ngram_range=(1,2),  
    max_features=5000,  
    sublinear_tf=True  
)
```

Feature Types

Unigrams

Examples:

research
study
climate
technology

Bigrams

Examples:

future research
machine learning
public health

10. CHARACTER-LEVEL TF-IDF FEATURES

A second feature space was introduced to capture stylistic writing patterns.

Objective

Capture:

- Writing style
 - Morphological patterns
 - Repeated token structures
-

Configuration

```
TfidfVectorizer(  
    analyzer='char_wb',  
    ngram_range=(3,5),  
    max_features=10000,  
    sublinear_tf=True  
)
```

Example Features

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ated

stud

Character-level features proved highly valuable during external validation because they captured stylistic signals independent of topic.

11. FEATURE UNION ARCHITECTURE

The final production solution combines both feature spaces.

Architecture

```
FeatureUnion([  
    ("word_tfidf"),  
    ("char_tfidf")  
])
```

Benefits

Word Features

Capture:

- Meaning
 - Vocabulary
 - Context
-

Character Features

Capture:

- Style
 - Formatting patterns
 - Construction habits
-

Combined Effect

Improved generalization on unseen content.

12. FEATURE SELECTION DECISIONS

The following table summarizes the final feature decisions.

Feature	Decision Reason
text_content	Retained Core prediction signal

Feature	Decision Reason
clean_text	Retained Standardized text
word_count	Removed Limited incremental value
avg_sentence_length	Removed Weak production impact
domain	Removed Not always available
topic_hint	Removed Leakage risk
source_model	Removed Direct label exposure
generation_method	Removed Not available during inference

13. FINAL PRODUCTION FEATURE SET

The final production pipeline uses:

Input

clean_text

Feature Layer

Word TF-IDF

- Unigrams
 - Bigrams
 - 5,000 Features
-

Character TF-IDF

- Character N-Grams (3–5)
 - 10,000 Features
-

Combined Feature Space

Approximately:

15,000 Features

14. FEATURE ENGINEERING IMPACT

The final feature architecture demonstrated:

Internal Validation

Metric Score

Accuracy 100%

Precision 100%

Recall 100%

F1 Score 100%

External Validation

Metric Score

Accuracy 82.05%

Precision 81.82%

Recall 91.67%

F1 Score 86.47%

The combination of Word TF-IDF and Character TF-IDF produced the strongest external validation performance among evaluated approaches.

15. RISKS & LIMITATIONS

Risks

- Vocabulary evolution
- New AI writing styles
- Domain drift

Mitigation

- Periodic retraining
- Additional validation datasets
- Ongoing monitoring

16. PROJECT CLOSURE STATEMENT

The feature engineering lifecycle successfully transformed raw textual content into a deployment-ready feature architecture optimized for AI content detection.

The final feature set balances predictive performance, deployment simplicity, leakage prevention, and generalization capability, making it suitable for production deployment by the client engineering team.

APPENDIX A – FINAL FEATURE FLOW

Raw Text

|



Cleaning

|



Word TF-IDF

|

|———▶ Unigrams

|

|———▶ Bigrams



Character TF-IDF

|

|———▶ 3–5 N-Grams



Feature Union



Model Input
